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RESEARCH ARTICLE

Light stress regulates pigment concentrations, growth, and gene expression in hatchery-cultivated *Kappaphycus alvarezii* (Solieriaceae, Rhodophyta)

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Abstract

Kappaphycus alvarezii is a commercially valuable red seaweed cultivated for biomass and phycocolloid production. Enhancing its productivity, stress tolerance, and phycocolloid yield requires genetic diversification and optimized cultivation techniques. This study investigated the effects of different light intensities on growth, pigment composition, and gene expression in wild *K. alvarezii* to inform hatchery cultivation strategies. Eight wild-type strains were clonally cultivated under low, medium (control), and high light conditions for 28 days. High light resulted in the highest daily growth rate (6.03% ± 1.86%), whereas phycoerythrin (PE) levels were significantly higher under low light conditions. Gene expression analysis revealed that high light upregulated antioxidant-related genes (e.g., carotenoid-associated genes) and genes involved in DNA synthesis and repair, while repressing photosynthesis-related genes, likely as a protective response against oxidative stress. Low light reduced the expression of light-harvesting-related genes and intensified the expression of polysaccharide-degradation genes, potentially compromising carrageenan biosynthesis. Notably, the lectin KAA protein was highly induced under low light, suggesting a potential protective role against mannose-dependent pathogens. These results highlight the acclimative mechanisms of *K. alvarezii* under light stress and provide insights into metabolic regulation and energy allocation. Understanding these responses can help inform seaweed farmers about optimal cultivation depth and light exposure to maximize growth and enhance carrageenan yield, thereby contributing to sustainable aquaculture practices. Furthermore, transcriptome profiling identified key genes associated with carrageenan biosynthesis; together with the identification of light-regulated processes, these results provide candidate molecular targets for future genotype-based improvement and cultivation strategies aimed at enhancing carrageenan production.

KEYWORDS
algae, carrageenan, gene expression, growth, lectin KAA, light intensity, pigments, Rhodophyta, stress response, wild *Kappaphycus alvarezii*

Abbreviations: ANOVA, analysis of variance; APC, allophycocyanin; BML, Bolinao Marine Laboratory; DEG, differentially expressed genes; DGR, daily growth rate; FDR, false discovery rate; GO, gene ontology; GSEA, Gene Set Enrichment Analysis; KEGG, Kyoto Encyclopedia of Genes and Genomes; LHC, light harvesting complex; ORA, over representation analysis; PAR, photosynthetically active radiation; PBP, phycobiliproteins; PC, phycocyanin; PDB, Protein Data Bank; PE, phycoerythrin; PFAM, protein families database; PGC, Philippine Genome Center; PSII, photosystem II; ROS, reactive oxygen species; UPMSI, Marine Science Institute University of the Philippines Diliman.

INTRODUCTION

Farmed *Kappaphycus alvarezii* biomass is an important commodity for carrageenan production in the aquaculture industry, particularly in tropical regions such as Indonesia and the Philippines (Borlongan et al., 2016; Reddy et al., 2017; Trono, 1992). Seaweed cultivation in the Philippines began in several regions in the late 1960s, with the Sulu Archipelago accounting for approximately half of the country's commercial production (Parker, 1974; Tahiluddin & Roleda, 2025).

Wild *Kappaphycus alvarezii* typically inhabits areas with slow to moderate water movement, attaching to sandy–coralline or rocky substrates from the lower intertidal to the upper subtidal zone (Trono, 1992). It has been observed growing in intertidal to subtidal habitats at water depths of less than 1 m and is sometimes partially exposed to desiccation during the lowest low tides, conditions characterized by extreme fluctuations in light intensity (Roleda et al., 2021). Wild populations of intertidal seaweeds, including *K. alvarezii*, have therefore developed adaptive and acclimative mechanisms that enable survival in highly variable environments. Fluctuating light intensity is a key selective pressure contributing to their physiological plasticity and persistence (Thien et al., 2021).

Light plays a crucial role in the growth and development of plants and algae, particularly red seaweeds of commercial importance such as eucheumatoids. Through photosynthesis, light energy is converted into chemical energy and stored as starch and carbohydrates, including carrageenan (Singh & Singh, 2015). Numerous studies have demonstrated that light intensity strongly influences photosynthetic efficiency and productivity in marine algae (Cleuvers & Ratte, 2002; Leukart & Lüning, 1994; Wimalasekera, 2019). Both insufficient and excessive light can negatively affect photosynthetic performance and other biological processes; however, algal responses to light intensity are often species-specific (Bialevich et al., 2022).

The primary and accessory pigments of red algae, including chlorophyll *a* (Chl *a*), carotenoids, and phycobiliproteins (PBPs), play essential roles in light harvesting and photoprotection (Chakdar & Pabbi, 2017; Osório et al., 2020; Pereira et al., 2012). The regulation of pigment-related gene expression by light intensity and quality has been investigated in microalgae such as dinoflagellates and diatoms (Shi et al., 2015; Valle et al., 2014), but remains poorly understood in commercially important macroalgae, including the eucheumatoid *Kappaphycus alvarezii*.

Seaweed farming has largely relied on a limited number of wild strains (novel haplotypes), with production primarily dependent on vegetative (clonal) propagation of a few commercially cultivated haplotypes. This practice, sustained for more than 60 years, has contributed to reduced genetic diversity in farmed eucheumatoid cultivars (Dumilag et al., 2023; Lim et al., 2014) and has been linked to increased susceptibility to pests and diseases, resulting in crop failures (Hurtado & Cheney, 2003). Consequently, most studies on *Kappaphycus alvarezii* have focused on growth rates and pigment concentrations of established commercial cultivars under varying light intensities or spectral conditions, often conducted in laboratory or farm settings (Aguirre-von-Wobeser et al., 2001; Borlongan et al., 2016; Indriatmoko et al., 2015). In contrast, the present study was conducted in a hatchery setting using newly collected wild strains and next-generation progeny of wild parental plants of *K. alvarezii*, as described by Hinaloc and Roleda (2021).

Previous studies have examined light-driven physiological and transcriptional responses in *Kappaphycus* and *Eucheuma*. For instance, Thien et al. (2021) investigated transcriptomic changes in *K. alvarezii* under different light wavelengths and CO_2 enrichment, and Liao et al. (2025) explored light-intensity adaptation and color variation in *Eucheuma perplexum*. These studies demonstrated that light conditions influence photosynthetic processes, pigment metabolism, and stress-related pathways. However, the molecular mechanisms linking light regulation to carbohydrate metabolism and extracellular polysaccharide biosynthesis in *K. alvarezii* remain largely uncharacterized. To address this knowledge gap, the present study investigated how varying light conditions influence transcriptional networks associated with carbon metabolism and cell-wall biosynthesis. Further molecular-level investigations are essential to deepen our understanding of how light intensity regulates gene expression, particularly genes involved in pigment biosynthesis, growth and development, photosynthesis, carbon fixation, stress responses, and carbohydrate metabolism, including floridean starch synthesis, carrageenan production, cellulose biosynthesis, and cell-wall formation in wild strains of *K. alvarezii*. Elucidating the molecular mechanisms underlying light responses in wild strains is crucial for comprehensively understanding their acclimative capacity, which is fundamental to optimizing growth and productivity under fluctuating light conditions.

MATERIALS AND METHODS

Collection, identification, and land-based hatchery experiment

Wild *Kappaphycus alvarezii* individuals were collected from a shallow reef flat (<1 m depth) in Guiuan, Samar, Philippines (10°56′45.6″ N, 125°44′05.7″ E). After removal of epibionts, samples were kept moist and cool during transport to the Marine Science Institute, University of the Philippines Diliman (UPMSI) in Quezon City for initial processing. The samples were

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subsequently transferred to the UPMSI Bolinao Marine Laboratory (UPMSI-BML) in Guiguiwanen, Bolinao, Pangasinan (16°22′41″ N, 119°54′42″ E) for hatchery cultivation. Species identity was molecularly confirmed following Roleda et al. (2021).

Vegetative fragments from individual plants were propagated separately in aquaria at the BML land-based hatchery. Reproductive individuals were induced to release spores, and the resulting progeny were reared separately to maturity, as described by Hinaloc and Roleda (2021). For the experiment, eight hatchery-cultivated strains of *Kappaphycus alvarezii* were used. Four strains were presumptive tetrasporophytes with a brown phenotype, originally collected from the wild, while the remaining four were tetrasporophyte progenies derived from cystocarpic female gametophytes and exhibited a green phenotype; all were collected from the same site. All samples were maintained under uniform cultivation conditions to minimize environmental variation and were clonally propagated to generate sufficient biomass for experimental replication.

Healthy, epiphyte-free branches from each selected *Kappaphycus alvarezii* strain were maintained under ambient hatchery conditions. All samples were initially cultured under medium light intensity (627 ± 64 μmol photons · m^{-2} · s^{-1}) for at least 1 year and clonally propagated for experimental use. For the experiment, fragments were transferred to 50-L glass aquaria (58 × 30 × 40 cm) supplied with a continuous flow of filtered seawater from the Guiguiwanen Channel and maintained under constant aeration. A total of 200–250 g of biomass from eight wild strains was randomly distributed among 16 aquaria and allowed to acclimate for 24 h prior to treatment.

The experimental setup consisted of two light regimes: (1) low light (203 ± 6 μmol photons · m^{-2} · s^{-1}), achieved by covering eight aquaria with two layers of black garden netting to filter approximately 77% of incident solar radiation and (2) high light (910 ± 23 μmol photons · m^{-2} · s^{-1}), established by placing the remaining eight aquaria in a fully sunlit area, with exposure to peak midday irradiance. Surface irradiance was measured using a LI-COR light meter (LI-14000, LI-COR, Nebraska, United States) equipped with a cosine sensor. Water temperature (≈31°C) was monitored daily using HOBO data loggers (UA-002–64, Onset Computer Corporation, Massachusetts, United States).

Growth rate

The growth rates of samples exposed to low and high light intensities were measured weekly over a 28-day period. Wet weights were recorded using a digital scale. Before weighing, samples were gently blotted with a clean cloth to remove excess surface moisture. Daily growth rates (DGR) were calculated following the method of Yong et al. (2013), using the formula:

$$\mathrm{DGR} = \left[\left(\frac{W_t}{W_o}\right)^{1/t} - 1\right] \times 100\,\%$$

where W_t = final weight in (g), W_o = initial weight (g), t = culture duration (days).

Pigment analyses

Sample collection for pigment extraction

Apical portions from samples exposed to low, medium (control), and high light intensities were collected for chlorophyll and phycobiliprotein analyses. Each treatment included eight replicates ($n = 8$) to ensure statistical reliability. Healthy apical fragments were rinsed with distilled water to remove surface contaminants and then stored at −80°C to preserve pigment integrity for subsequent analysis.

Chlorophyll and phycobiliprotein determination

Samples from the medium light (control) treatment were collected prior to the 28-day light exposure experiment, whereas samples from the low and high light treatments were collected at the end of the 28-day period. The experiment was conducted between May and June in Bolinao, Pangasinan, Philippines, where average temperatures ranged from 28 to 31°C, with approximately 11.3 h of daily sunshine in May and 10 h in June.

Chlorophyll *a* and phycobiliproteins (PBPs) were extracted following modified protocols described by Narvarte et al. (2022). All procedures were carried out in a dark room at 16°C to minimize pigment degradation. Frozen apical portions were ground in liquid nitrogen using a mortar and pestle. For Chl *a* extraction, 10 mL of absolute ethanol (100%) was added to each powdered sample, whereas for PBP extraction, 10 mL of 1.0 M phosphate buffer (pH 7.6) was used. Extractions were performed in separate 15-mL polyethylene centrifuge tubes and incubated overnight at 4°C. Samples were then centrifuged at 11,176 *g* for 15 min to separate the pigment-containing supernatant.

Absorbance was measured using a microplate spectrophotometer (Epoch 2, BioTek, Vermont, United States). Chlorophyll *a* absorbance was recorded at 632, 649, 665, and 696 nm, whereas PBP absorbance was measured at 562, 615, and 652 nm. Pigment concentrations were calculated using the equations described by Ritchie (2008) and Bennett and Bogorad (1973).

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$$\text{Chl } a\left(\mu g \cdot g^{-1}\ \text{FW}\right) = \left[\frac{\left(0.064\left(A_{632}\right)\right) - \left(4.5224\left(A_{649}\right)\right) + \left(13.2969\left(A_{665}\right)\right) - \left(1.7453\left(A_{696}\right)\right)}{\text{FW}}\right] \times V$$

$$\text{PC}\left(\mu g \cdot g^{-1}\ \text{FW}\right) = \left[\frac{\left(A_{615}\right) - \left(0.474\left(A_{652}\right)\right)}{5.34}\right] \frac{V}{FW} \times 1000$$

$$\text{APC}\left(\mu g \cdot g^{-1}\ \text{FW}\right) = \left[\frac{\left(A_{652}\right) - \left(0.208\left(A_{615}\right)\right)}{5.09}\right] \frac{V}{FW} \times 1000$$

$$\text{PE}\left(\mu g \cdot g^{-1}\ \text{FW}\right) = \left[\frac{\left(A_{652}\right) - \left(2.41(\text{PC})\right) - \left(0.849(\text{APC})\right)}{9.62}\right] \frac{V}{FW} \times 1000$$

where PC represents phycocyanin, APC for allophycocyanin, and PE for phycoerythrin, which are all expressed as $\mu g \cdot g^{-1}$ fresh weight (FW), A_x represents absorbance at the specified wavelength, and V for volume of extraction solvent (mL).

Statistical analyses

All results have been presented as mean ± standard deviation (*SD*). A one-way analysis of variance (ANOVA) was used to assess differences in DGR and pigment concentrations among the three irradiance treatments (high, medium—the control—and low light), with a sample size of $n = 8$. Post hoc comparisons were performed using the Bonferroni test to determine statistically significant differences ($p < 0.05$). Statistical analyses were conducted using SPSS version 23 (IBM Corp., Armonk, New York, United States).

RNA extraction and sequencing

Apical portions of thalli were collected from healthy, epiphyte-free samples under medium (control, $n = 8$), high ($n = 8$), and low ($n = 8$) light conditions, yielding a total of 24 samples for RNA sequencing. Medium light samples were collected prior to the 28-day light exposure experiment and served as the Day 0 baseline for transcriptome analyses, whereas low and high light samples were collected after 28 days of exposure. Physiological measurements were conducted across all treatments, including the medium light control, at all sampling time points to confirm treatment-specific responses.

All samples were briefly rinsed with autoclaved seawater to remove surface contaminants, followed by a rapid rinse with distilled water to eliminate residual salts that could interfere with RNA extraction. Exposure to distilled water was minimized and applied uniformly across treatments to reduce potential osmotic stress effects; thus, any transient stress response, if present, was consistent among samples and unlikely to bias comparative transcriptomic analyses. Tissue samples were then cut into smaller fragments using a sterile blade, transferred to sterile 5-mL microcentrifuge tubes, sealed with parafilm, and immediately flash-frozen in liquid nitrogen using a Dewar canister.

Frozen samples were ground into a fine powder using a sterile mortar and pestle under continuous liquid nitrogen cooling. Approximately 350 mg of powdered tissue was transferred to sterile, precooled 2-mL microcentrifuge tubes. Total RNA was extracted using Ambion TRIzol reagent and further purified with the Zymo RNA Clean & Concentrator-5 kit. RNA purity was assessed using a NanoDrop 2000 spectrophotometer (Thermo Scientific, United States), with acceptable A260/A280 ratios of approximately 2.0, and RNA integrity was confirmed by gel electrophoresis. All RNA samples underwent additional quality control at the Philippine Genome Center (PGC) using a MultiNA system (Shimadzu) and a Qubit Fluorometer (Invitrogen). Library preparation was performed using the TruSeq mRNA Library Prep Kit (Illumina), followed by next-generation sequencing on a NextSeq 500/550 platform (2 × 75 cycles) using the High Output v2.5 kit at PGC, Diliman, Quezon City, Philippines.

Gene expression analysis

RNA sequencing was performed on an Illumina NextSeq 500 platform using TruSeq Stranded mRNA library preparation (2 × 75 cycles). All libraries fell within the expected size range of 260–300 bp and had concentrations ranging from 12.3 to 35.9 $ng \cdot \mu L^{-1}$. Quality control of raw sequence data was conducted using FastQC (Andrews, 2010), and the preprocessing and filtering of reads were performed using Trimmomatic (Bolger et al., 2014). FastQC quality metrics were visualized using the ngsReports package. Principal component analysis (PCA) was applied to identify patterns in global gene expression.

Transcript abundance was quantified using Kallisto, which calculated read counts from paired-end FASTQ files by mapping reads to the *Kappaphycus alvarezii* reference transcriptome generated in this study. Differential gene expression analysis was conducted in the R environment using edgeR (Robinson et al., 2010). Lowly expressed genes were filtered out using a threshold of counts per million (CPM) > 1 in at least two samples. The Benjamini–Hochberg procedure was applied

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Parameter	Low light	Medium/control light	High light
% Daily growth rate (DGR · day^{-1})	1.88 ± 1.38^{a}	4.94 ± 1.55^{ab}	6.03 ± 1.86^{b}
Phycoerythrin conc (μg · g^{-1} FW)	6.52 ± 2.52^{a}	2.30 ± 1.40^{b}	3.19 ± 2.00^{b}
Phycocyanin conc (μg · g^{-1} FW)	2.49 ± 1.25	1.69 ± 0.34	3.03 ± 1.47
Allophycocyanin conc (μg · g^{-1} FW)	3.24 ± 1.26	2.14 ± 0.59	3.71 ± 2.11
Chlorophyll *a* conc (μg · g^{-1} FW)	1207 ± 691	723 ± 354	1334 ± 459

Note: Values are means ± *SD* ($n = 8$). Different superscript letters within a row indicate significant differences at $p < 0.05$. Parameters without superscripts (PC, APC, Chl *a*) did not show statistically significant differences among light treatments.

TABLE 1 Mean percent daily growth rate (%DGR), and pigment concentrations of wild strains of *Kappaphycus alvarezii* after exposure to different light intensities.

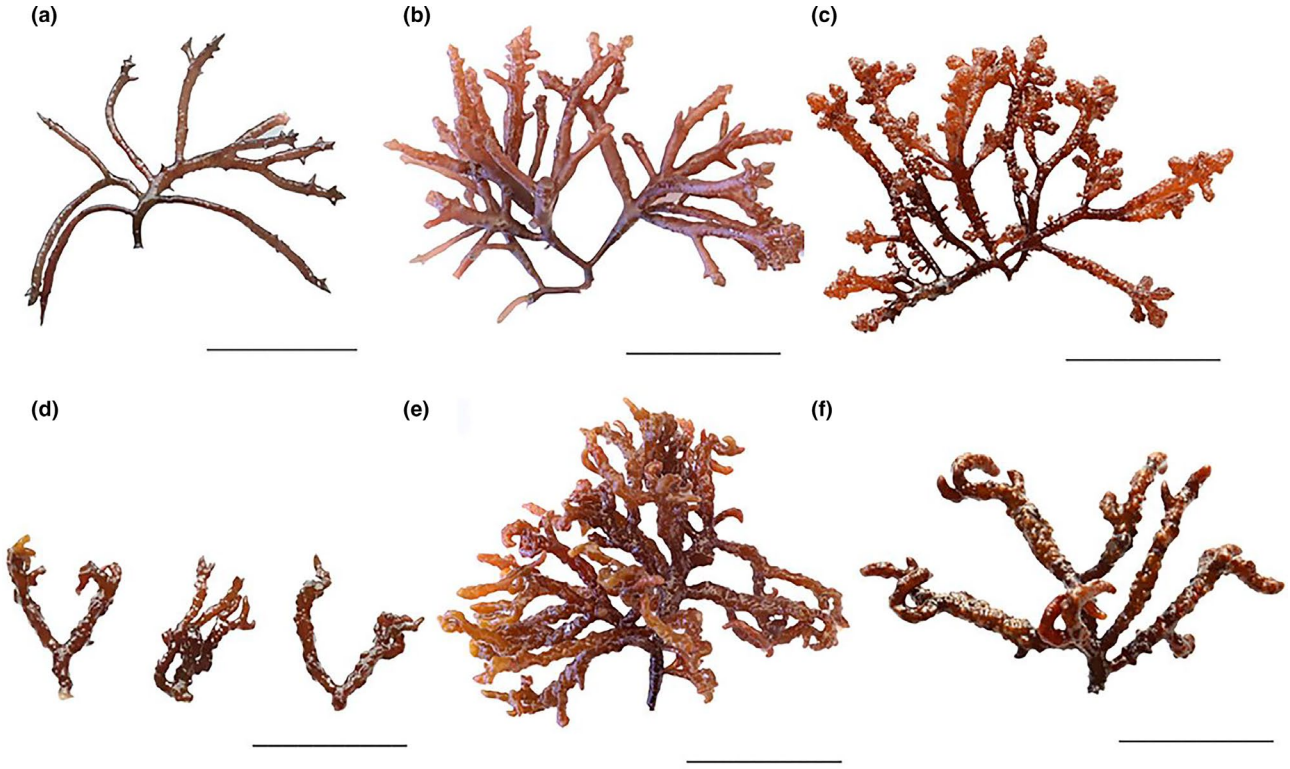


FIGURE 1 Representative samples of wild strains of brown color phenotype of presumptive tetrasporophytic *Kappaphycus alvarezii* used in experiment representing replicates no. 2 (a–c) and no. 3 (d–f) under low (a and d), medium = control (b and e) and high (c and f) light intensities. The specimen under medium light was clonally propagated for >1 year while specimen under low and high light intensities were after 28 days experimental exposure. Scale bar = 10 cm.

to adjust *p*-values for multiple testing, and genes with a false discovery rate (FDR) < 0.05 and an absolute $\log_2$ fold change ($\log_2$FC) > 1.5 were considered significantly differentially expressed.

Functional and pathway enrichment analyses were performed using the clusterProfiler package (Yu et al., 2012) to associate differentially expressed genes (DEGs) with biological processes, molecular functions, and metabolic pathways. The DEG annotations were subjected to over representation analysis (ORA; Yu et al., 2012) and Gene Set Enrichment Analysis (GSEA; Subramanian et al., 2005), focusing on gene ontology (GO) categories (Biological Process, Molecular Function, Cellular Component), protein families database (PFAM) protein domains, and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways. Results from enrichment analyses were compiled and visualized using ShinyKaGe for RNA-seq Data Analysis (v2.3), developed by Fahrenkrug et al. (2023). The KEGG pathways were further visualized and annotated with differential expression scores using the pathview package (Luo & Brouwer, 2013).

Biological samples were deposited in the NCBI BioSample database under accession numbers SAMN35040676–SAMN35040699, and raw sequencing reads are available in the NCBI Sequence Read Archive (SRA) under BioProject ID PRJNA971596, with run accessions SRR24507657–SRR24507680. The transcriptome assembly was deposited as a Transcriptome Shotgun Assembly (TSA) at DDBJ/ENA/GenBank under accession GLKY00000000; the version described in this study corresponds to GLKY01000000.

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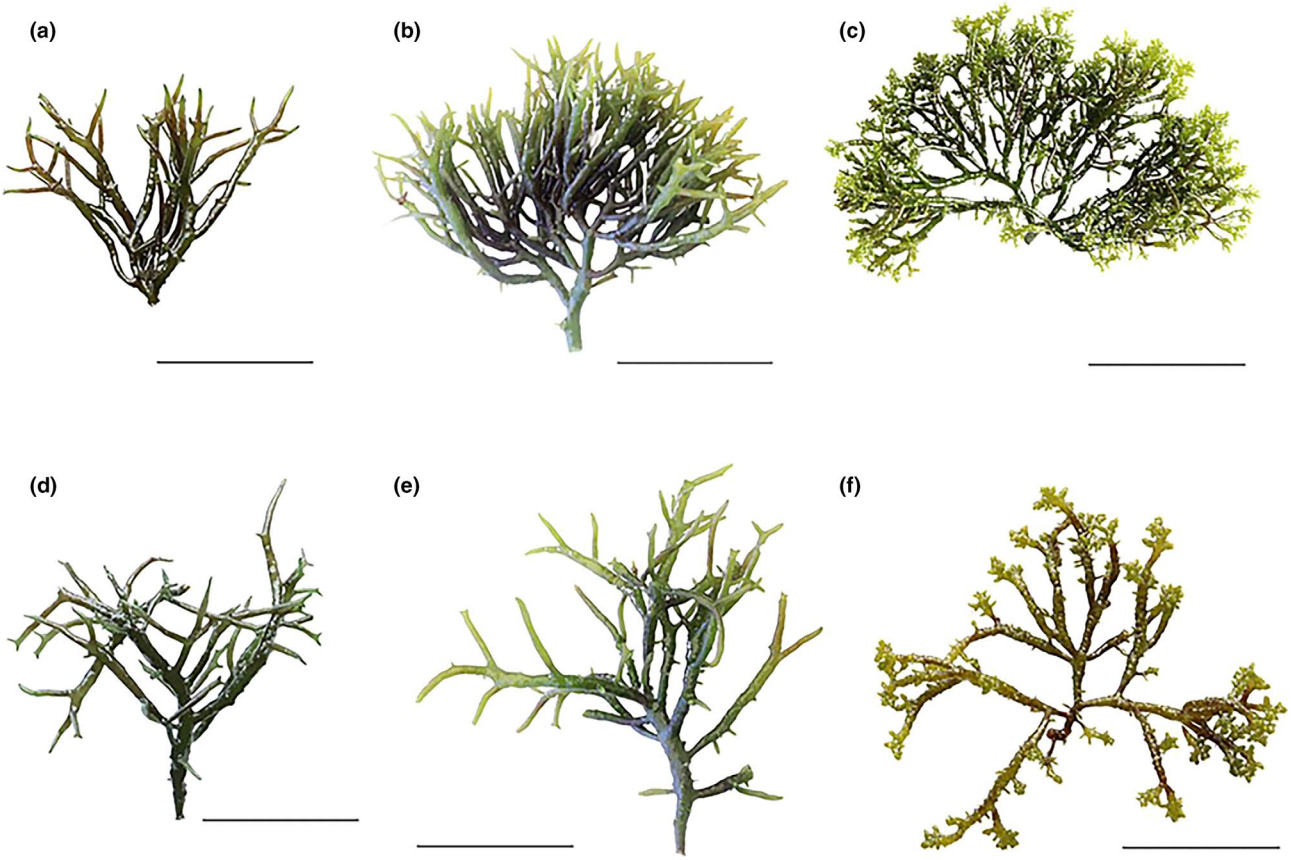


FIGURE 2 Representative samples of tetrasporophyte progenies exhibiting green color phenotype derived from cystocarpic wild parental plants of *Kappaphycus alvarezii* used in the experiment representing replicate no. 1 (a–c) and no. 3 (d–f) under low (a and d), medium = control (b and e) and high (c and f) light intensities. The specimen under medium light was clonally propagated for >1 year while specimen under low and high light intensities were after 28 days experimental exposure. Scale bar = 10 cm.

RESULTS

Growth rate

The DGR of wild *Kappaphycus alvarezii* strains was monitored weekly for 28 days under high, medium (control), and low light conditions in an outdoor land-based hatchery. As shown in Table 1, samples exposed to high light exhibited the highest mean DGR ($6.03\% \pm 1.86\% \cdot day^{-1}$), followed by those under medium light ($4.94\% \pm 1.55\% \cdot day^{-1}$), whereas the lowest growth was observed under low light conditions ($1.88 \pm 1.38\% \cdot day^{-1}$). One-way ANOVA revealed a significant effect on the DGR ($F_{2,21} = 14.23$, $p = 0.0001$, $F_{crit} = 3.47$). Bonferroni-corrected pairwise comparisons indicated significant differences between low and medium light treatments ($p = 0.001$) and between low and high light treatments ($p = 0.0002$), whereas no significant difference was detected between medium and high light treatments ($p = 0.23$).

Morphological differences were also evident among treatments (Figures 1 and 2). Samples transferred from medium to high light developed bushier, more highly branched thalli with lighter-colored new growth, whereas samples maintained under low light produced thinner, darker thalli. Prolonged exposure to low light resulted in thallus softening and tissue deterioration (Figure 3).

Pigment concentrations

Pigments are essential for light absorption during photosynthesis. This study evaluated the effect of varying light intensities on the concentrations of primary (Chl *a*) and accessory pigments such as PE, PC, APC in *Kappaphycus alvarezii*. Pigment levels were measured before and after the 28-day light exposure experiment. As shown in Table 1, mean concentrations of Chl *a* ($\mu g \cdot g^{-1}$ FW) did not significantly differ among light treatments, as indicated by one-way ANOVA ($F_{2,21} = 0.87$, $p = 0.43$, $F_{crit} = 3.47$). The PBPs were measured before and after light exposure. All samples were initially grown under medium light (control) before being transferred to low or high light conditions. The PE concentration significantly increased under low light ($6.52 \pm 2.52\,\mu g \cdot g^{-1}$ FW). A one-way ANOVA showed that light intensity significantly affected the concentration of PE ($F_{2,21} = 9.61$, $p = 0.001$,

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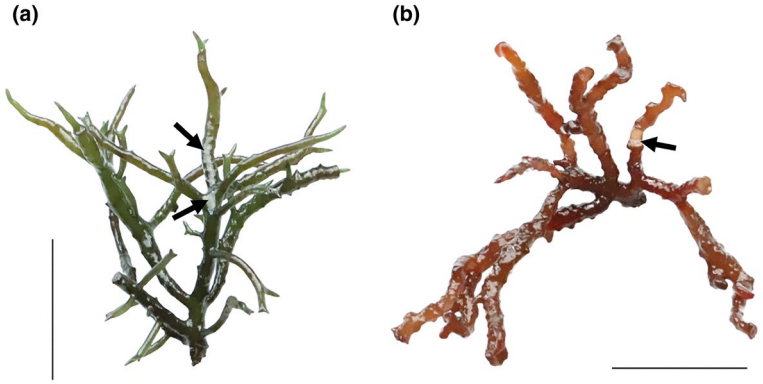


FIGURE 3 Thallus degeneration after 28 days cultivation under low light intensity. Green (a) and brown (b) color phenotypes of *Kappaphycus alvarezii* exhibiting bleaching, softening, and necrosis pointed by arrows. Scale bar = 10 cm.

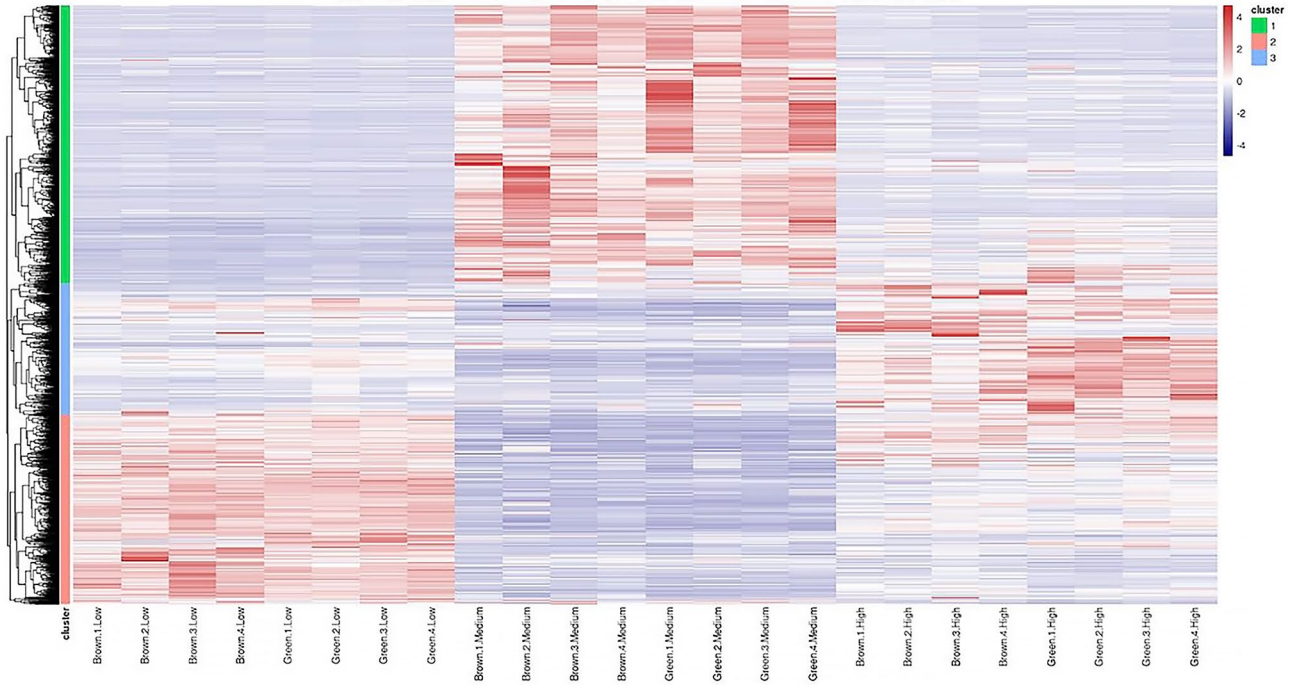


FIGURE 4 Heat map analysis of differential gene expression in wild strains of *Kappahycus alvarezii* under varying light intensities. Cluster labels are displayed in the upper right corner, with cluster coverage shown on the *y*-axis. The *x*-axis represents individual samples under specific light intensity treatments. The color gradient in the upper right indicates the $\log_2$FC for each gene, with red signifying upregulated genes, (+) and blue signifying downregulated genes (–). Three distinct clusters (green, light pink, light blue) were observed.

$F_{crit} = 3.47$). Bonferroni post hoc analysis revealed significant differences between low and high light ($p = 0.011$) and between low and medium light ($p = 0.001$), whereas no significant difference was detected between high and medium light treatments ($p = 0.32$). In contrast, one-way ANOVA showed no significant effect on PC and APC across light treatments ($F_{2,21} = 2.83$, $p = 0.08$, $F_{crit} = 3.47$, and $F_{2,21} = 2.57$, $p = 0.09$, $F_{crit} = 3.47$, respectively).

Gene expression

RNA sequencing was performed on the Illumina NextSeq 500 platform using TruSeq Stranded mRNA library preparation (2 × 75 cycles). Across all samples, library sizes fell within the expected range of 260–300 bp, with concentrations ranging from 12.3 to 35.9 ng · μL^{-1}. Pooled libraries were diluted to a final loading concentration of 1.6 pM. Sequencing of 24 samples generated approximately 431 million paired-end reads (~18 million reads per sample), yielding a total of ~64.6 Gb of data. The sequencing run achieved a cluster density of 203 K · mm^{-2}, with 81.9% of clusters passing filter, indicating high sequencing quality. The average percentage of bases with Q ≥ 30 was 88.7%, confirming high sequencing accuracy.

Principal component analysis was conducted to assess variation in gene expression among light treatments (Figure S1). The first two principal components explained 67.9% of the total variance (PC1 = 52.4%, PC2 = 15.5%) and clearly separated samples according to light intensity. The strong separation along PC1

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TABLE 2 Differentially expressed genes associated with biofilm formation, identified through KEGG pathway analysis using ORA and GSEA.

Gene name	Gene ID	High vs. low log_2FC	High vs. medium log_2FC	Low vs. medium log_2FC	ips_PANTHER_desc
Cellulose Synthase Catalytic Subunit A [UDP-Forming]	KAG.1_10131	2.69	0.21 (N/S)	−2.48	Cellulose Synthase Catalytic Subunit A [UDP-Forming]
Family GT2	KAG.1_23692	2.05	−0.09 (N/S)	−2.13	Cellulose Synthase Catalytic Subunit A [UDP-Forming]
Cellulose Synthase Catalytic Subunit A [UDP-Forming]	KAG.1_24791	2.69	0.04 (N/S)	−2.65	Cellulose Synthase Catalytic Subunit A [UDP-Forming]
Acyl coenzyme A diacylglycerol acyltransferase	KAG.1_3169	2.07	−0.02 (N/S)	−2.10	Cellulose Synthase Catalytic Subunit A [UDP-Forming]
Family GT2	KAG.1_9767	2.94	1.68	−1.28 (N/S)	Cellulose Synthase Catalytic Subunit A [UDP-Forming]

Note: Log_2 fold change (log_2FC) values >1.5 indicate significant differential expression, with positive values (+) representing upregulation and negative values (−) representing downregulation.

Abbreviation: N/S, non-significant results.

indicates that light intensity was the primary driver of global transcriptional variation. Samples from low, medium, and high light treatments formed distinct clusters, reflecting consistent biological responses to light conditions. Subtle sub-clustering within the medium and high light groups suggests that color phenotype (green vs. brown) also contributed to transcriptional variation, indicating potential differences in light response between morphotypes.

Of the 9905 expressed genes, 2251 were identified as differentially expressed ($log_2FC > 1.5$, $FDR < 0.05$, adjusted $p \leq 0.05$). The complete list of DEGs is provided in Table S1. Hierarchical clustering of DEGs revealed distinct expression profiles among wild *Kappaphycus alvarezii* strains grown under low, medium, and high light intensities (Figure 4).

Cluster 1 (Figure 4, green) comprised 1043 genes that were downregulated under both low and high light but upregulated under medium light. These genes were enriched in pathways associated with photosynthesis, plastid organization, and isoprenoid metabolism, with corresponding enrichment in KEGG pathways related to carbon fixation, starch metabolism, and carotenoid (isoprenoid) biosynthesis. Notably, genes involved in cellulose biosynthesis, particularly those associated with sugar polymerization, were downregulated under low light relative to medium and high light conditions (Table 2).

Cluster 2 (Figure 4, pink) contained 712 genes that were upregulated under low light and downregulated under medium and high light. Although no significant enrichment was detected for biological processes or cellular components, molecular function analysis revealed significant enrichment for mannose-binding activity. This was supported by PFAM domain analysis showing enrichment of OAA-family lectin sugar binding domains (Table 3; Figure 5).

Cluster 3 (Figure 4, light blue) included 496 genes that were upregulated under high light but downregulated under low and medium light. These genes were enriched for functions related to DNA replication, DNA repair, biosynthesis, chromatin organization, mitosis, cell-cycle progression, and chromosome-associated components (Figure 6a). Gene Set Enrichment Analysis further revealed that genes associated with fructose-1,6-bisphosphatase, the reductive pentose phosphate cycle, photosynthesis (dark reactions), and carbon fixation were downregulated under high light relative to medium light conditions (Figure 6b).

Detailed pairwise comparison of the effect of light on gene expression

A total of 603 genes were differentially expressed between high and low light conditions ($log_2FC > 1.5$, $FDR < 0.05$, adjusted $p < 0.05$). Notably, chlorophyll *a*/*b* binding proteins (Gene IDs: KAG.1_7618, KAG.1_32,002, KAG.1_5665, KAG.1_1064, KAG.1_27968, KAG.1_19757) exhibited the highest expression ($log_2FC > 6$) under high light. These genes were enriched in GO terms, KEGG pathways, and PFAM domains, as confirmed by both ORA and GSEA. Most annotations were associated with photosynthesis, light harvesting, photoprotection (e.g., antioxidant activity such as peroxidases), and isoprenoid biosynthesis (carotenoids; Table 4).

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TABLE 3 Differential expression of genes annotated with the OAA-family lectin sugar binding domain (SBD), enriched in PFAM domains via overrepresentation analysis.

Gene names	Gene ID	HvsL log_2FC	HvsM log_2FC	LvM log_2FC	ips_ pfams	ips_IPs_ desc	UniProt_ homolog species	UniProt_ homolog GeneID
Lectin ESA-2, Lectin KSA2, Lectin KAA-1, Lectin KSL	KAG.1_27951	−1.21 (N/S)	0.91 (N/S)	2.13	SBD	OAA-family lectin sugar binding domain	*Eucheuma serra, Kappaphycus striatus, Kappaphycus alvarezii*	KSA2, KAA-1, KSL
Lectin KAA-1, Lectin ASL-2	KAG.1_27952	−3.31	−0.25 (N/S)	3.06	SBD	OAA-family lectin sugar binding domain	*Kappaphycus alvarezii*	KAA-1
Lectin KAA-2	KAG.1_28018	−1.14 (N/S)	1.19 (N/S)	2.34	SBD	OAA-family lectin sugar binding domain	*Kappaphycus alvarezii*	KAA-2
Lectin KAA-2	KAG.1_32173	−1.01 (N/S)	0.90 (N/S)	1.92	SBD	OAA-family lectin sugar binding domain	*Kappaphycus alvarezii*	KAA-2
Lectin ASL-2, Lectin KAA-1	KAG.1_32186	−3.33	−0.09 (N/S)	3.24	SBD	OAA-family lectin sugar binding domain	*Agardhiella subulata*	ASL-2
Lectin KAA-1, Lectin KSA2, Lectin KSL	KAG.1_32191	−1.16 (N/S)	1.45 (N/S)	2.62	SBD	OAA-family lectin sugar binding domain	*Kappaphycus striatus*	KSA2, KSL
Lectin MPA-2, Lectin ASL-2	KAG.1_28019	NA	NA	NA	SBD	OAA-family lectin sugar binding domain	*Meristotheca papulosa, Agardhiella subulata*	MPA-2, ASL-2
Lectin KSA2, Lectin KAA-1, Lectin EDA-2	KAG.1_32170	N/A	NA	NA	SBD	OAA-family lectin sugar binding domain	*Kappaphycus alvarezii*	KAA-1
Lectin KSA2, Lectin KAA-1, Lectin KAA-2	KAG.1_32171	0.05 (N/S)	−2.90	−2.92	SBD	OAA-family lectin sugar binding domain	*Kappaphycus striatus, Kappaphycus alvarezii*	KSA2, KAA-1

Note: Log_2 fold change (log_2FC) values are shown for high versus low (HvL), high versus medium (HvM), and low versus medium (LvM) light conditions. Values >1.5 indicate significant differential expression, with positive values (+) representing upregulation and negative values (–) representing downregulation.

Abbreviation: N/S, non-significant results.

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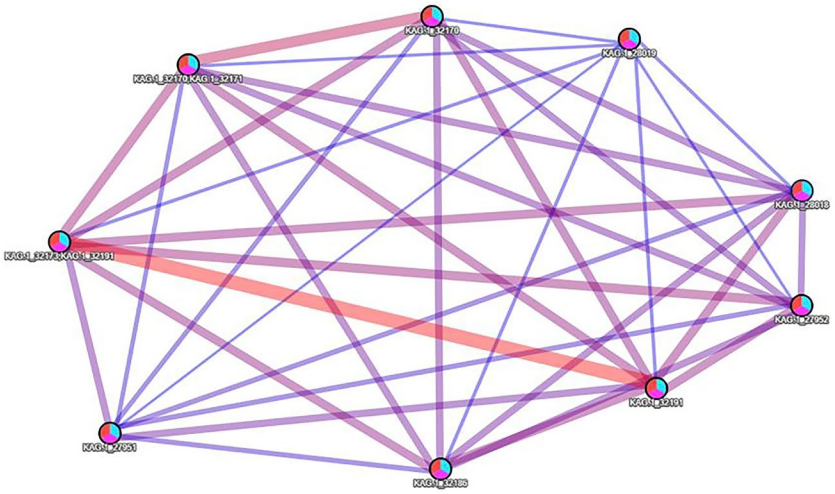


FIGURE 5 Markov cluster algorithm (MCL) plot showing genes associated with lectin KAA homologs. This plot visualizes gene relationships based on sequence similarity. Each node (circle) represents a gene associated with lectin KAA homologs. Edges (lines) indicate similarity between genes; thicker or differently colored edges reflect stronger similarity. Genes are grouped into clusters based on their connectivity, with each cluster representing a potential gene family, functional module, or orthologous group.

The KEGG pathway analysis (Figure 7a) revealed functional clustering, including an intriguing association between O-antigen nucleotide sugar biosynthesis and photosynthesis, as well as starch and sucrose metabolism. Genes involved in both O-antigen biosynthesis and photosynthesis were upregulated (Table S2), suggesting a functional link between carbon fixation and cell-wall remodeling. High light conditions appeared to redirect carbon toward sugar nucleotides, which are critical for cell-wall biosynthesis, potentially strengthening structural integrity in response to environmental stress. PFAM domain analysis (Figure 7b) showed enrichment of genes associated with chlorophyll *a-b* binding, An-peroxidases (KAG.1_18399, KAG.1_18406, KAG.1_18427, KAG.1_18430, KAG.1_30684, KAG.1_3156), glycosyltransferases, fatty acid desaturases, phycobilisome (PBS) linkers, hydrolases, NAD-binding proteins, and aminotransferases, all upregulated under high light. The upregulation of these genes likely reflects acclimation mechanisms enhancing light absorption, energy transfer, reactive oxygen species (ROS) scavenging, cell-wall remodeling, and stabilization of light harvesting complexes. Conversely, zinc finger proteins (zf-RING2) were downregulated, suggesting a shift in regulatory priorities—reducing transcriptional repression to favor photoprotection and stress acclimation. Reduced ubiquitin-mediated protein turnover may also reflect a transition from proteolytic to non-proteolytic strategies under high light.

A total of 1236 genes were differentially expressed between high and medium light ($\log_2$FC > 1.5, FDR < 0.05, adjusted $p < 0.05$). The DNA replication licensing factor MCM5 exhibited the highest upregulation under high light ($\log_2$FC = 4.8) and was identified in both ORA and GSEA analyses. Genes enriched across GO, KEGG pathways, and PFAM domains included protein kinase domains and fructose-1,6-bisphosphatase (a bifunctional protein), which were down-regulated under high light. This downregulation suggests an energy trade-off under high light, prioritizing homeostasis and photoprotection over carbon storage. Under high light, genes involved in DNA replication, repair, and antioxidant activity were upregulated compared to medium light (Figure 8a), including bromoperoxidase and superoxide dismutase. Conversely, genes associated with photosynthesis, carbon fixation, and carbohydrate metabolism were down-regulated (Figure 8b), indicating a regulatory mechanism to limit ROS overproduction.

A total of 1840 genes were differentially expressed between low and medium light ($\log_2$FC > 1.5, FDR < 0.05). The glycine cleavage system P protein (KAG.1_16041) showed the most significant downregulation ($\log_2$FC = −9.5) and was enriched in both ORA and GSEA analyses. This reduction under low light suggests minimized photorespiratory flux and C_1 metabolism as an adaptive strategy to conserve energy and maintain metabolic homeostasis. Other enriched genes included protein kinase domains, fructose-1,6-bisphosphatase, and chlorophyll *a-b* binding protein 1B-21 (chloroplastic). Most genes involved in photosynthesis-antenna proteins (Table 5), carbon fixation, and carbohydrate metabolism were down-regulated under low light compared to medium light (Figure 9), reflecting disrupted metabolic processes and altered energy allocation. Consistently, low light samples exhibited reduced growth rates and thinner thalli, indicating energy stress that may impair normal growth and development.

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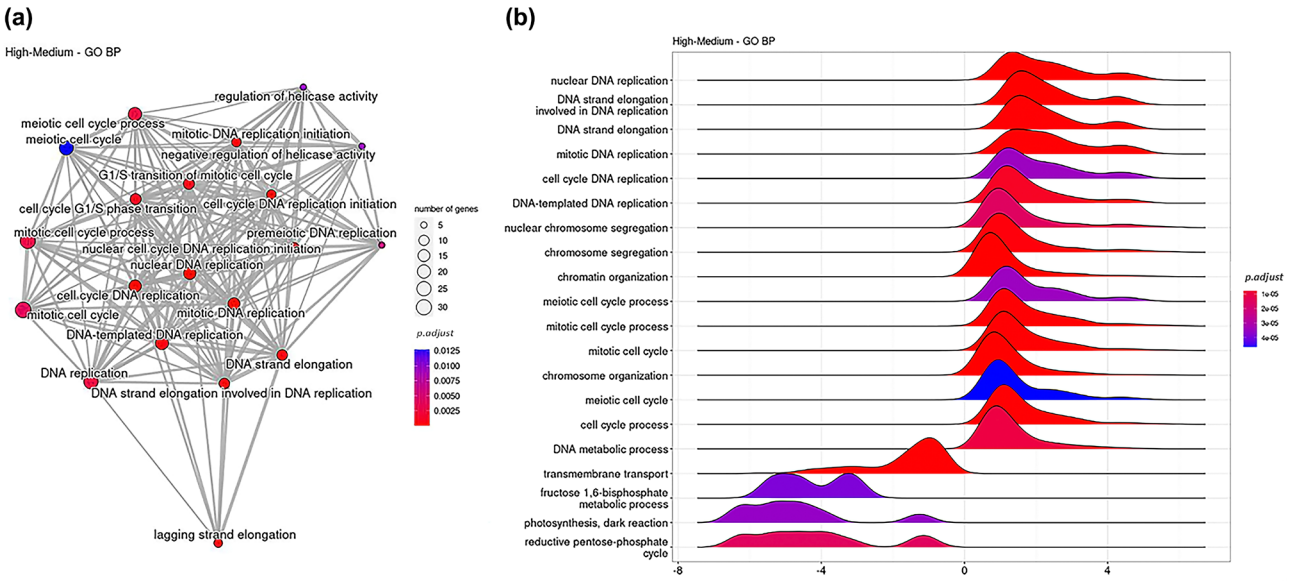


FIGURE 6 Enrichment map plot (a) displaying the differentially expressed genes (DEGs) identified by ORA for GO biological processes in high versus medium light in Cluster 3. In the enrichment map plot, the nodes (circles) show enriched gene sets, the size of the node shows the number of genes/belonging to a specific gene set, the color gradient indicates the *p*-adjusted value, the smaller the *p*-value the higher is the enrichment significance, while the edges (connections) link pathways that have shared genes, forming clusters of related functions. Thicker edges mean higher similarity between gene sets (more shared genes). Clusters with many thick edges suggest functionally related processes. Sparse connections suggest distinct biological themes. Ridge plot (b) showing enriched genes identified by GSEA for GO biological processes in high versus medium light condition. The *y*-axis shows enriched processes, the *x*-axis represents $\log_2$FC (+ value: upregulated, – value: downregulated), and the color gradient reflects the *p*-adjusted value. The ridge shapes indicate how gene-level scores vary: narrow and peaked curves indicate tighter gene score distributions, while wider curves suggest more variability.

Dynamic expression of genes involved with carbohydrate flux

To gain deeper insight into carbohydrate-related functions, the predicted proteome of *Kappaphycus alvarezii* was analyzed using the CAZy database via dbCAN2 (updated to dbCAN3) to identify proteins involved in carbohydrate metabolism. Candidate genes were annotated based on sequence homology, structural comparisons with proteins in the Protein Data Bank (PDB), and manual curation to identify genes potentially involved in carrageenan biosynthesis. Genes associated with light harvesting, carbon fixation, and photosynthesis were included in the analysis only if they were predicted to interact with carbohydrates, ensuring a focused investigation of carbohydrate-related functions. A less stringent threshold ($\log_2$FC > 1.0, FDR > 0.05) was applied to enhance sensitivity in detecting DEGs. Among CAZy and predicted carbohydrate-flux genes, a total of 268 DEGs were identified and grouped into three clusters based on expression patterns across high, medium, and low light conditions (Figure 10).

Cluster 1 comprised 43 DEGs that were strongly upregulated under low light and moderately upregulated under high light relative to medium light. Notably, xylulose kinase (KAG.1_17511) showed the highest expression under low light and elevated expression under high light. Most DEGs in this cluster were associated with carbohydrate metabolism and sugar processing, including enzymes such as phosphoglucomutase and kinases (e.g., alpha-glucan water dikinase). Polysaccharide degradation genes were also upregulated under low light, including glycoside hydrolases (GHs) degrading cellulose (endoglucanase, cellobiohydrolase, beta-glucosidase) and carrageenan (e.g., beta-galactosidase). Lytic polysaccharide monooxygenase (CAZy: AA15 family), which degrades cellulose, chitin, and hemicellulose, was significantly upregulated under high light but down-regulated under low light. The increased expression under high light suggests active cell-wall remodeling in response to elevated ROS and redox activity, whereas reduced expression under low light likely reflects energy conservation. This trade-off may limit growth under low light, as cell-wall remodeling contributes to thallus expansion.

Cluster 2 contained 135 DEGs predominantly downregulated under both low and high light relative to medium light. The sugar phosphate transporter domain-containing protein (KAG.1_29016) showed the greatest downregulation under low light ($\log_2$FC –7.3) and high light ($\log_2$FC –4.3), suggesting reduced carbohydrate transport under light stress to maintain energy balance. Ribulose-5-phosphate-3-epimerase (KAG.1_7205) was strongly downregulated under high light ($\log_2$FC –6.3), consistent with decreased Calvin cycle activity and a strategic allocation of carbon toward cell-wall remodeling or

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TABLE 4 Isoprenoid biosynthesis-related genes, particularly those associated with carotenoid biosynthesis, showing significant upregulation under different light conditions.

Gene name	Gene ID	KEGG pathway node description	High vs. low light (log_2FC)	High vs. medium light (log_2FC)	Low vs. medium light (log_2FC)
Lycopene beta-cyclase	KAG.1_12184	Lycopene beta-cyclase [EC:5.5.1.19]	4.59	1.48 (N/S)	−3.1
15-cis-phytoene synthase	KAG.1_1342	15-cis-phytoene synthase [EC:2.5.1.32]	2.43	−0.64 (N/S)	−3.08
Beta-carotene hydroxylase	KAG.1_3275	Beta-carotene hydroxylase [EC:1.14.13.-]	1.86	−1.37 (N/S)	−3.23
Geranylgeranyl pyrophosphate synthase, chloroplastic	KAG.1_7012	Geranylgeranyl diphosphate synthase, type I [EC:2.5.1.1, 2.5.1.10, 2.5.1.29]	2.19	−1.40 (N/S)	−3.60
1-deoxy-D-xylulose-5-phosphate synthase	KAG.1_17528	1-deoxy-D-xylulose-5-phosphate synthase [EC:2.2.1.7]	2.74	0.07 (N/S)	−2.66

Note: Log_2 fold change (log_2FC) values >1.5 indicate significant differential expression; positive values (+) represent upregulation, negative values (−) represent downregulation.

Abbreviation: N/S, non-significant results.

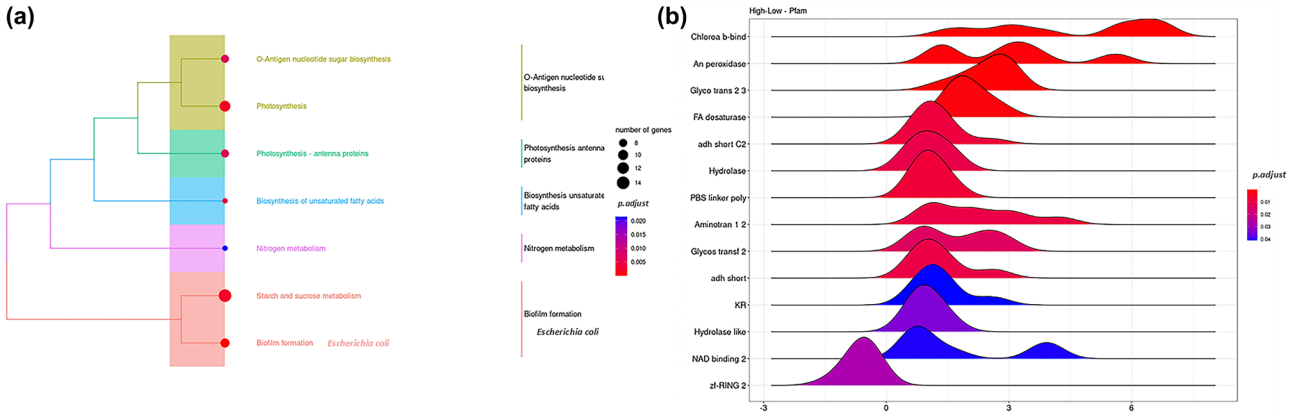


FIGURE 7 Tree plot (a) of significantly enriched KEGG pathways in the high versus low light comparison. Each branch groups pathways with similar gene enrichment profiles. Colored rectangles indicate functional clusters. Dot size represents the number of enriched genes per pathway, while dot color reflects the adjusted *p*-value (*p*.adjust), with red indicating higher significance. Ridge plot (b) showing enriched genes identified by PFAM domains.

antioxidant defense. Other downregulated genes included those involved in starch biosynthesis (granule-bound starch synthase 1) and glycosylation (mannosyltransferase), as well as carrageenan biosynthesis genes (Table 6). Under low light, the downregulation of photosynthesis-related genes, chloroplastic transporters, lipid and galactolipid biosynthesis, and nucleotide sugar metabolism (e.g. PAPST, sugar phosphate transporters) indicates prioritization of energy conservation over structural growth and polysaccharide sulfation. Notably, lectin KAA was upregulated under low light, suggesting a protective role against mannose-dependent pathogens, potentially compensating for reduced polysaccharide biosynthesis.

Cluster 3 included 90 DEGs upregulated under high light but downregulated under low light relative to medium light. Phosphopyruvate hydratase (KAG.1_16585) exhibited the highest expression under both high (log_2FC 5.8) and low (log_2FC 5.1) light, indicating a role as a metabolic checkpoint to maintain energy homeostasis during light stress. High light upregulation may meet ATP demands for photoprotective processes, while low light upregulation supports essential metabolism under limited photosynthetic output. Antioxidant-related genes, including 15-cis-phytoene synthase (log_2FC 2.4) and ascorbate peroxidase (log_2FC 1.9), were elevated under high light. Conversely, low light caused significant downregulation of genes involved in energy production (ATP synthase, log_2FC −1.8) and cell-wall biosynthesis (cellulose synthase catalytic subunit A, log_2FC −2.4). Under high light, carbohydrate-modifying enzymes such as xyloglucan endotransglucosylase (log_2FC 1.3) were upregulated, suggesting metabolic

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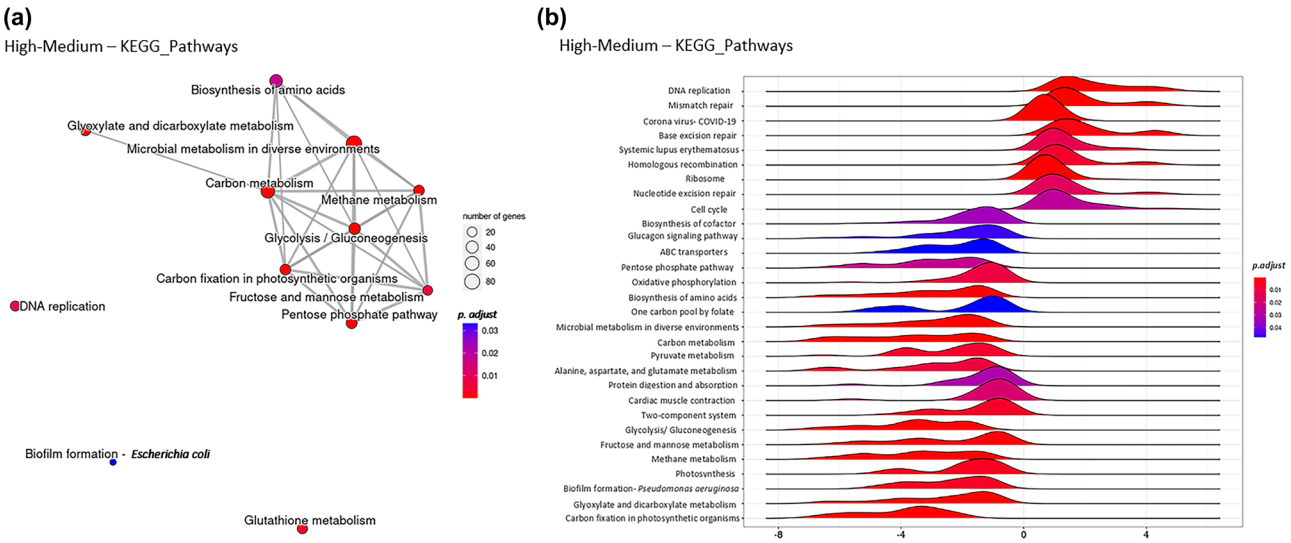


FIGURE 8 Enrichment map plot (a) and ridge plot (b) displaying processes with detected enrichment in KEGG pathways in high versus medium light conditions.

reprogramming and cell-wall reinforcement as adaptive responses to light stress.

DISCUSSION

Growth and pigment responses under light stress

Wild strains of *Kappaphycus alvarezii* exhibited pronounced light-dependent physiological responses. The highest daily growth rate observed under 910 μmol photons · m^{-2} · s^{-1} suggests efficient photoprotection as the first line of defense against high irradiance. This likely involves downregulation of photosystem II (PSII), where excess energy absorbed by chlorophyll, but not used in photosynthesis, is dissipated as heat. Photoinhibition occurs only when PSII is inactivated or damaged (Buma et al., 2024). This result aligns with Borlongan et al. (2016), who reported that *K. alvarezii* did not exhibit photoinhibition—a reduction in effective photochemical quantum yield (F_v/F_m)—even under 1000 μmol photons · m^{-2} · s^{-1} of photosynthetically active radiation (PAR).

Despite energetic and physiological trade-offs, such as investments in photoprotective pigments (carotenoids, xanthophylls, polyphenols), DNA repair systems, and enzymatic antioxidants, the high growth rates observed here (6.03% ± 1.86% · day^{-1}) indicate coordinated photoprotection with net positive productivity. These growth rates surpassed those reported for commercial cultivars in field conditions—2.5% · day^{-1}, 3.1%–3.3% · day^{-1}, and ~5% · day^{-1} (Doty, 1987; Kasim & Mustafa, 2017; Wakibia et al., 2006)—highlighting the superior physiological potential of novel wild strains. Variability in field-grown cultivars may also reflect seasonal changes, environmental fluctuations, and grazing losses.

Morphological variations were also evident: High light induced increased branching and lighter coloration, whereas low light led to darker, thinner thalli and, in some cases, tissue deterioration or erosion, characteristic of energy-limited and stress-induced softening (Lee et al., 2019). Light intensity strongly influenced pigment composition, particularly PE, which was significantly elevated under low light. The increase in PE, coupled with stable PC and APC levels, suggests chromatic acclimation to optimize photon capture. Similar low light induction of PE has been reported in the microalgal cryptophyte *Rhodomonas salina* and rhodophytes *Porphyridium* spp. (Jahn et al., 1984; Xie et al., 2021), as well as an ~10% increase in PE-based light-harvesting capacity in the coralline macroalga *Lithothamnion glaciale* (= *Boreolithothamnion glaciale*; Voerman et al., 2022). Flexibility in phycobiliprotein synthesis allows red algae to maintain photosynthetic efficiency under fluctuating light (Cao et al., 2021; Guan et al., 2013; Talarico & Maranzana, 2000). These results emphasize that pigment reorganization is an early and efficient mechanism of light acclimation in *Kappaphycus alvarezii*.

Transcriptomic responses to high and low light

Principal component analysis of the transcriptome (RNA-Seq) data revealed distinct clustering of samples according to light intensity (high, medium, and low), indicating that light intensity is a major determinant of gene expression in wild strains of *Kappaphycus alvarezii*. These clusters reflected shared transcriptomic profiles shaped by light conditions, consistent with previous

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TABLE 5 Differential expression of light harvesting and bilin biosynthesis genes in *Kappaphycus alvarezii* under different light intensities.

Gene name	Gene ID	KEGG pathway node description	Differential expression in high vs. low light (log_2FC)	Differential expression in high vs. medium light (log_2FC)	Differential expression in low vs. medium light (log_2FC)
PBS-linker domain-containing protein	KAG.1_6158	Phycocyanin-associated rod linker protein (cpcC)	1.61	0.12 (N/S)	–1.48 (N/S)
Phycobilisome 31.8kD linker polypeptide	KAG.1_13566	Phycoerythrin-Associated Linker Protein (Cpec)	1.42 (N/S)	2.31	0.88 (N/S)
Bilin Biosynthesis Protein	KAG.1_11226	Bilin Biosynthesis Protein (CpeU, CpeZ)	0.10 (N/S)	–2.76	–2.88
Chlorophyll a-b binding protein, chloroplastic	KAG.1_1064	Light Harvesting Complex I Chlorophyll A/B Binding Protein 1 (Lhca1)	6.62	–0.91 (N/S)	–7.55
Chlorophyll A-B binding protein	19,757	Light Harvesting Complex I Chlorophyll A/B Binding Protein 4 (Lhca4)	3.89	–0.67 (N/S)	–4.57
Chlorophyll a-b binding protein 1B-21, chloroplastic	3753	Light Harvesting Complex I Chlorophyll A/B Binding Protein 5 (Lhca5)	1.44 (N/S)	–1.12 (N/S)	–2.57
Chlorophyll a-b binding protein 7, chloroplastic	5665	Light Harvesting Complex II Chlorophyll A/B Binding Protein 3 (Lhcb3, Lhcb7)	5.75	–0.79 (N/S)	–6.56
Light harvesting complex I polypeptide	32,002	Light Harvesting Complex II Chlorophyll A/B Binding Protein 5 (Lhcb5, Lhcb6)	6.38	–0.47 (N/S)	–6.86

Note: Positive values indicate upregulation, while negative values indicate downregulation; KEGG Pathway Node Descriptions: Provide insights into the functional roles of the genes in metabolic processes.
Abbreviation: N/S, no significant change in expression.

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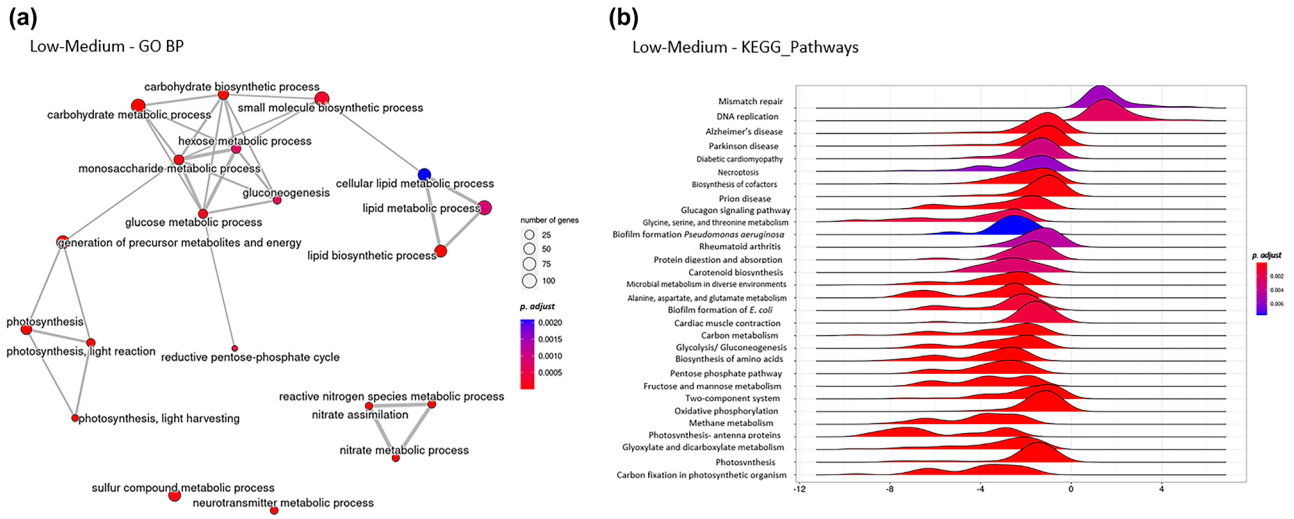


FIGURE 9 Tree plot (a) and ridge plot (b) of significantly enriched KEGG pathways in the low versus medium light comparison.

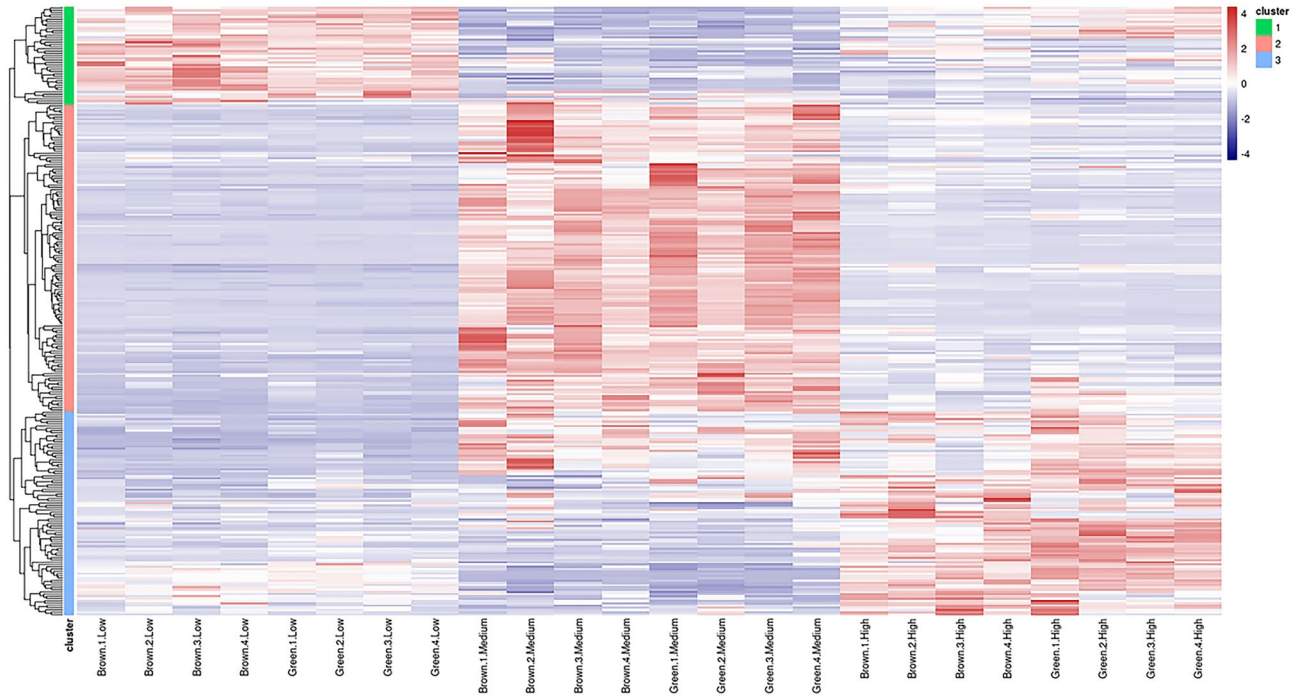


FIGURE 10 Heatmap of differentially expressed genes (DEGs) involved in carbohydrate flux in *Kappaphycus alvarezii* with $\log_2$FC values >1.0, FDR < 0.05, and *p*-adjusted value < 0.05. The genes were grouped into three clusters based on their expression profiles, with each cluster represented by a distinct color as shown in the upper right legend. The color gradient indicates the $\log_2$FC, with red denoting upregulated genes and blue representing down-regulated genes. The horizontal axis represents the individual samples of wild strains of *K. alvarezii*.

reports that red algae can adapt to variations in both light intensity and quality (Talarico & Maranzana, 2000). Although light intensity accounted for most of the variance, residual variation may be attributed to factors such as color phenotype, time of collection, and other variables. Our study, however, focused primarily on the effects of light intensity. Previous work by Thien et al. (2021) demonstrated transcriptomic responses to light wavelength and CO_2 in *K. alvarezii*. Comparison with our results suggests that light intensity is the primary regulator of growth, pigment composition, and global gene expression in wild populations. By integrating physiological, pigment, and transcriptomic data, our results indicate that light intensity controls carbon allocation, cell-wall remodeling, and carbohydrate flux. Similarly, Liao et al. (2025) examined pigment-based photoadaptation in *Eucheuma perplexum*, whereas our study has highlighted broader transcriptional regulation of growth and structural metabolism in wild *K. alvarezii* strains under variable light conditions.

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TABLE 6 Differentially expressed genes associated with carrageenan biosynthesis.

Gene name	Gene ID	High vs. low log_2FC	High vs. medium log_2FC	Low vs. medium log_2FC
Glycosyltransferase	KAG.1_401	0.62 (N/S)	−4.70	−5.32
Carbohydrate sulfotransferase	KAG.1_26153	1.42	−3.14	−4.57
Carbohydrate sulfotransferase	KAG.1_27916	4.40	0.66 (N/S)	−3.75
Carbohydrate sulfotransferase	KAG.1_22911	1.94	−1.50	−3.43
Glycosyltransferase 2-like domain-containing protein	KAG.1_33618	3.05	−0.23 (N/S)	−3.28
Carbohydrate sulfotransferase	KAG.1_33196	2.00	−1.13	−3.13
Carbohydrate sulfotransferase	KAG.1_27918	0.60 (N/S)	−2.12	−2.74
Carbohydrate sulfotransferase	KAG.1_27915	1.70	−0.93 (N/S)	−2.63
Carbohydrate sulfotransferase	KAG.1_33192	1.14	−1.09	−2.23
UDP-glucose 4-epimerase	KAG.1_459	0.59 (N/S)	−0.91 (N/S)	−1.50
Glycosyl transferase, family GT8	KAG.1_4375	0.71 (N/S)	−0.54 (N/S)	−1.25
UDP-glucose 4-epimerase	KAG.1_17166	0.36 (N/S)	−0.89 (N/S)	−1.25

Note: $Log_2FC > 1.0$ shows significant expression, with positive values (+) representing upregulation and negative values (−) representing downregulation.
Abbreviation: N/S, non-significant results.

High irradiance elicited a coordinated protective response in *Kappaphycus alvarezii*, prioritizing photoprotection, redox homeostasis, and cellular maintenance. Upregulation of light harvesting proteins, carotenoid biosynthesis pathways, and antioxidant enzymes under high light aligned with known photoprotective strategies in *Fucus* spp. and other photosynthetic organisms, in which excess excitation energy necessitates enhanced oxidative stress mitigation (Pearson et al., 2009; Squires et al., 2019; Stadnichuk & Krasilnikov, 2024; Vranová et al., 2012). Increased expression of carotenoid biosynthesis genes, including those involved in lycopene cyclization, likely contributes to ROS quenching and thylakoid membrane stabilization under elevated irradiance (Kirilovsky & Kerfeld, 2013). Concurrently, activation of ascorbate peroxidase-linked processes and terpenoid backbone biosynthesis reinforces redox-regulatory capacity, supporting sustained cellular function during light-induced oxidative stress (Jalal et al., 2023; Rezayian et al., 2019). Collectively, these responses reflected an integrated acclimation strategy enabling *K. alvarezii* to maintain photosynthetic integrity and cellular stability under excessive light.

In low light conditions, PBP expression increased, potentially as a compensatory mechanism for light capture. In the cryptophyte *Rhodomonas salina*, a lack of correlation between pigment concentrations and gene expression suggested that pigment regulation is largely post-transcriptional (Schomaker et al., 2023). Nevertheless, this mechanism did not fully sustain photosynthetic efficiency in *Kappaphycus alvarezii*. Reduced expression of photosynthesis-related genes under low light triggered downstream effects, including impaired carbon fixation and reduced biosynthesis of polysaccharides essential for cell-wall formation, ultimately hindering growth and causing thallus degradation. Phenotypic and transcriptomic analyses suggest that under low light, the alga reallocates internal resources for survival, effectively consuming its reserves to maintain essential metabolic functions.

Under high light, *Kappaphycus alvarezii* balanced growth promotion with stress mitigation. Upregulation of DNA replication licensing factors, such as MCM5, and genes involved in genome maintenance suggests enhanced cell division and repair processes to counter oxidative stress (Hihara et al., 2001; Li et al., 2022; Muramatsu & Hihara, 2012; Roleda et al., 2007). Simultaneously, downregulation of photosynthetic and carbon fixation pathways—including RuBisCO and fructose-1,6-bisphosphatase—may limit ROS overproduction. This apparent disconnect between transcriptional activity and physiological output likely reflects post-transcriptional regulation, as seen in *Chlamydomonas reinhardtii*, in which chloroplast RNA-binding proteins modulate mRNA stability and translation in response to light (Rochaix, 1996). Overall, these results reveal a flexible strategy allowing *K. alvarezii* to sustain cellular function and growth under high light while shifting toward energy conservation under low light. Future studies linking gene expression to physiological traits will provide further insight into these adaptive mechanisms (Feder & Walser, 2005; Maier et al., 2009).

Carbohydrate flux and energy reallocation

Light intensity significantly influenced carbohydrate metabolism, highlighting the metabolic flexibility of *Kappaphycus alvarezii*. Genes involved in carbohydrate utilization and degradation were upregulated under low light, including those encoding phosphoglucomutase and alpha-glucan water dikinase. This pattern reflected a metabolic shift toward mobilizing stored carbohydrates to sustain energy production when

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photosynthesis is limited, similar to stress responses observed in *Chlamydomonas reinhardtii* and *Chlorella vulgaris* (Blifernez-Klassen et al., 2012; Zachleder & Brányiková, 2013). These results suggest that *K. alvarezii* breaks down stored carbohydrates, such as floridean starch, into glucose to maintain glycolysis and basal metabolic activity under low light stress.

Conversely, high light conditions also likely promoted starch degradation, but in this context to regulate carbon supply and support energy-demanding processes—including DNA replication, nuclear division, protein synthesis, and cytokinesis—necessary to maintain cellular function and genomic integrity during proliferation and oxidative stress (Ishihara et al., 2022). In contrast, genes related to carbon fixation, starch biosynthesis, and glycosylation (e.g., mannosyltransferases) were down-regulated under both low and high light, indicating a trade-off between growth-related biosynthesis and survival metabolism.

Under high light, although growth was promoted, carbon fixation genes such as ribulose-5-phosphate 3-epimerase were downregulated, accompanied by upregulation of glycolytic and antioxidant pathways, reflecting a shift from carbon storage toward protective metabolism. Lytic polysaccharide monooxygenase (AA15 family), upregulated in high light but repressed under low light, likely contributes to cell-wall remodeling and redox activity associated with ROS. The expansion of this gene family in *Kappaphycus alvarezii* may represent an adaptive trait for maintaining structural resilience during light stress. Collectively, these transcriptional adjustments reveal a dual strategy: mobilization of energy reserves under low light and redirection of carbon flux toward photoprotection and cell wall reinforcement under high light.

Regulation of carrageenan biosynthesis and lectin KAA expression

Both high and low light intensities resulted in reduced expression of genes related to carrageenan biosynthesis and glycosyltransferases, suggesting that light stress alters carbon partitioning away from polysaccharide production. Under low light, this suppression likely reflects a metabolic shift toward carbohydrate degradation and energy conservation, whereas under high light, it may correspond to a redirection of carbon toward pathways associated with antioxidant defense and cell wall remodeling.

Notably, the mannose-binding lectin KAA was strongly upregulated under low light. Lectins are carbohydrate-binding proteins involved in stress signaling, defense, and cell-wall modulation (Chrispeels, 1991; Hwang & Hwang, 2011). KAA-2, a high-mannose-specific lectin from *Kappaphycus alvarezii*, has exhibited potent antiviral activity by binding viral glycoproteins (Hirayama et al., 2016; Rodrigues et al., 2025). The marked upregulation of KAA under stress suggests that this lectin may contribute to defense responses, particularly given its strong affinity for mannose-decorated pathogens. Such activity could provide a protective advantage against opportunistic bacteria, including *Vibrio* species, highlighting a promising avenue for applied research. Strains that constitutively express KAA, or that can be selectively bred for elevated lectin production, may exhibit improved resilience against microbial infections. Although additional replication studies are needed to verify the stress-induced expression patterns observed here, these results underscore the potential of KAA as a functional biomarker and a candidate for enhancing disease resistance in commercially cultivated *K. alvarezii*. The coordinated regulation of lectins and cell wall-associated polysaccharide genes points to an integrated response for maintaining structural integrity under stress. Although we did not perform biochemical quantification of carrageenan, the transcriptional evidence supports the hypothesis that *K. alvarezii* adjusts its extracellular polysaccharide composition in response to light availability. Future studies combining transcriptomics with direct biochemical assays are needed to confirm these patterns.

Our results also showed upregulation of glycoside hydrolases, accompanied by concurrent downregulation of glycosyltransferases and genes involved in carrageenan and starch biosynthesis under low light. This pattern indicated a metabolic shift from polysaccharide synthesis toward degradation. The elevated expression of glycoside hydrolases suggests active breakdown of structural and storage polysaccharides, whereas the reduced expression of glycosyltransferases points to decreased synthesis of these molecules. Together, these changes imply a net loss of polysaccharides, likely as part of an energy conservation strategy and reallocation of carbon resources under stress. It is important to note that the annotations of glycosyltransferases and glycoside hydrolases in this study were primarily based on homology and structural conservation of proteins in the PDB from 15 algal species. As such, these results should be considered as identifying potential candidate genes rather than definitive functional assignments.

CONCLUSIONS

This study underscored the significant impact of light intensity on the physiological and molecular responses of wild *Kappaphycus alvarezii*, particularly in growth performance, pigment concentration, and gene expression. High light exposure (~910 μmol photons · m^{-2} · s^{-1}) promoted higher growth rates, likely supported by enhanced photoprotection, including increased antioxidant activity, regulated photosynthesis,

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and upregulated mitosis, protein synthesis, and DNA replication and repair mechanisms. In contrast, low light conditions suppressed growth and induced thallus fragmentation, suggesting energy limitation, and a shift toward degradation of stored carbohydrates such as cellulose and starch to sustain metabolic demands. Transcriptomic analysis revealed dynamic regulation of genes involved in photosynthesis, carbon fixation, and carbohydrate metabolism under both light extremes. Some genes related to pigments showed significant differential expression, particularly chlorophyll *a*-*b* binding proteins; however, some genes associated with phycobiliproteins, especially PE, showed low correlation with the obtained pigment concentrations, which may indicate that their regulation may occur predominantly at the post-transcriptional level. Low light stress notably downregulated genes involved in carrageenan biosynthesis and glycosylation, possibly due to limited UDP-glucose availability for galactose synthesis. Concurrent upregulation of lectin KAA genes suggests a protective mechanism that may influence carrageenan structure and cell-wall remodeling. Under high light, selective downregulation of photosynthesis and carbon-fixation genes appeared to serve as a photoprotective strategy to reduce ROS generation, despite overall metabolic activation that supported increased growth. Both low and high light intensities were associated with distinct alterations in carbohydrate flux, with low light potentially limiting carbon allocation for growth and high light driving a reallocation of carbohydrates that coincided with enhanced growth, as well as cell-wall structure and phycocolloid production. The results obtained from this study provided an insight on how *K. alvarezii* responds to molecular and physiological processes after exposure to different light intensities. Our study revealed that key genes linked to carbohydrate metabolism, stress response, and lectin activity showed different patterns of expression, revealing the molecular pathways that shape how the seaweed survive and maintain growth under light stress. These results lead to possible practical applications in seaweed cultivation. Depending on the target product and goal of cultivating *K. alvarezii*, our study has offered an insight on possible use of careful light adjustments during hatchery and grow-out stages, which could aid in boosting biomass production, improving carrageenan quality, increasing floridean starch accumulation, and other valuable products like lectin, by steering metabolism in beneficial ways. Overall, the results indicate that maintaining appropriate light intensities can enhance biomass production in *K. alvarezii*, whereas the identification of key light-responsive genes highlights important molecular targets underlying light tolerance and growth regulation. These candidate genes provide a foundation for future genetic selection, strain improvement, and cultivation strategies aimed at enhancing productivity and resilience under variable light conditions.

AUTHOR CONTRIBUTIONS

April Mae Tabonda-Nabor: Conceptualization (equal); data curation (lead); formal analysis (lead); investigation (lead); methodology (lead); resources (equal); validation (lead); visualization (lead); writing – original draft (lead); writing – review and editing (lead). **Bea A. Crisostomo:** Conceptualization (supporting); data curation (supporting); formal analysis (supporting); investigation (supporting); methodology (equal); resources (supporting); validation (supporting); writing – review and editing (supporting). **Alexandros Armaos:** Data curation (equal); formal analysis (equal); software (equal); validation (equal); visualization (equal). **Gabriel Schweizer:** Data curation (equal); formal analysis (equal); software (equal); validation (equal); visualization (equal). **Arturo O. Lluisma:** Data curation (supporting); investigation (supporting); methodology (supporting); resources (supporting); supervision (supporting); validation (supporting); writing – review and editing (supporting). **Michael Y. Roleda:** Conceptualization (equal); funding acquisition (lead); investigation (equal); methodology (equal); project administration (lead); resources (equal); supervision (lead); validation (supporting); writing – review and editing (equal). **Scott C. Fahrenkrug:** Conceptualization (equal); data curation (equal); formal analysis (equal); funding acquisition (lead); investigation (supporting); methodology (equal); project administration (equal); resources (equal); software (lead); supervision (lead); validation (equal); visualization (equal); writing – original draft (equal); writing – review and editing (equal).

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(OVPAA-BPhD-2019-06) to MYRoleda titled Nutrient physiology of commercially important eucheumatoids: two-phase cultivation strategy to optimize biomass and carrageenan yield and quality. The project provided resources for the completion of the Master's thesis of AMTabonda-Nabor titled Influence of light stress on growth, pigment concentrations, and gene expression in wild strains of *Kappaphycus alvarezii* (Solieriaceae, Rhodophyta).

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DATA AVAILABILITY STATEMENT

Data will be made available upon request from the authors.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Principal component analysis (PCA) plot showing the clustering of biological replicates of *Kapphycus alvarezii* cultured under low (green), medium (blue), and high (red) light intensities. Each point represents an individual sample, and the shaded ellipses indicate the 95% confidence interval for each light treatment group. The first two principal components (PC1 and PC2) explain 52.4% and 15.5% of the total variance, respectively. Distinct clustering among the three light groups demonstrates clear transcriptional differentiation in response to light intensity, with high reproducibility among biological replicates within each treatment.

Table S1. Complete list of 2251 differentially expressed genes (DEGs) identified in *Kappaphycus alvarezii* under varying light conditions. Columns include gene name, gene ID, log2 fold change, log CPM.

Table S2. Differentially expressed genes associated with O-antigen nucleotide sugar biosynthesis and photosynthesis in different light conditions. Log_2FC values >1.5 shows significant expression. Positive values indicate upregulation and negative values denote downregulation.

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